# The AI chip supply chain in brief

The fifteen stages that turn sand and software into an AI accelerator: who leads each one, how concentrated it is, and where China and the United States stand.

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**Concentration** counts the firms that credibly supply a stage. Every stage rates Extreme or High.

**Substitutability** rates how hard it would be for a well-funded state or firm to build an alternative: Very hard, Hard or Moderately hard.

| # | Stage | Who leads | Concentration | Substitutability |
|---|---|---|---|---|
| 01 | Chip Design, EDA and IP | Synopsys US, Cadence US | ▮ Extreme | Hard |
| 02 | Silicon and Wafers | Shin-Etsu Handotai JP, SUMCO JP | ▫ High | Moderate |
| 03 | Chemicals, Gases and Photoresist | JSR JP, Tokyo Ohka Kogyo JP | ▫ High | Moderate |
| 04 | Lithography | ASML NL, Nikon JP | ▮ Extreme | Very hard |
| 05 | Photomasks and Pellicles | AGC JP, Hoya JP | ▮ Extreme | Hard |
| 06 | Deposition and Etch | Applied Materials US, Lam Research US | ▫ High | Moderate |
| 07 | Metrology and Inspection | KLA US, Applied Materials US | ▫ High | Moderate |
| 08 | Transistors and the Front End | TSMC TW, Intel US | ▮ Extreme | Hard |
| 09 | Foundries and Fabs | TSMC TW, Intel Foundry US | ▮ Extreme | Hard |
| 10 | Memory and HBM | SK hynix KR, Samsung KR | ▮ Extreme | Hard |
| 11 | Advanced Packaging | TSMC TW, Intel US | ▮ Extreme | Moderate |
| 12 | Substrates and PCBs | Ajinomoto JP, Nitto Boseki JP | ▮ Extreme | Hard |
| 13 | Test and Assembly | Advantest JP, Teradyne US | ▫ High | Moderate |
| 14 | Systems and Networking | Nvidia US, Broadcom US | ▫ High | Moderate |
| 15 | Data Centers and Power | GE Vernova US, Hitachi Energy CH | ▫ High | Moderate |

## Across the chain

Two chapters look across all fifteen stages, at what a chip costs and at the contest over who controls them.

### 16 The Economics of AI Chips

Epoch AI estimated in December 2025 that Nvidia's B200 AI chip cost about $6,400 to build, mostly for memory and packaging. Its processor silicon cost about $900.

Epoch AI says Nvidia's margin on Blackwell may be lower than the gap between the B200's build cost and its selling price suggests. It cites reports from March 2024 that put the B200's selling price at $30,000 to $40,000 per chip. Epoch AI notes that most Blackwell revenue comes from servers and rack-scale systems, which may carry lower margins.

Sources: chipsupplychain.org/component-costs/

### 17 Who Controls the Chain

Since 2018 the United States has used export controls to limit China's access to advanced chips and chipmaking tools, and China has been learning to make both.

Japan and the Netherlands have aligned their own export controls with the American rules on chipmaking equipment. China has answered by building its own tools and by restricting sales of US chips in China. Chinese toolmakers supplied 35 percent of China's own chipmaking equipment market in 2025, up from 10 to 15 percent before 2022. Nvidia says the Chinese government restricted the sales of its H200 chip that the United States licensed in February 2026.

Sources: chipsupplychain.org/geopolitics/

# The fifteen stages

In the order a chip is made.

## 01 Chip Design, EDA and IP

**Extreme concentration** · Hard to substitute

Synopsys, Cadence and Siemens EDA make the software used to design every advanced chip. In 2025 they took over 85 percent of its $18 billion market, licensed blocks included.

- **Synopsys** `United States`. 90%+ of the timing-signoff market at advanced nodes as of 2026
- **Cadence** `United States`. 55-60% of the hardware emulator market at advanced nodes as of 2026

**China** China designs its own AI accelerators, but Huawei's most powerful chip in 2026, the Ascend 950, delivers about half the computing performance of Nvidia's H100 from 2022, on Epoch AI's estimate. CSIS found that as of 2025 Chinese design tools were still far from replacing the full range of advanced capabilities US firms offer, especially for 5 nm and 3 nm chips.

**United States** Synopsys and Cadence, the two largest design-software vendors by 2025 revenue, are American.

Sources: chipsupplychain.org/design-and-eda/

## 02 Silicon and Wafers

High concentration · Moderately hard to substitute

Firms based in Japan, Taiwan, Germany and South Korea make most 300 mm wafers used for advanced chips. China makes most polysilicon, almost all of it for solar cells.

- **Shin-Etsu Handotai** `Japan`. Its parent's electronics materials segment, wafers included, had ¥1,015.7bn of sales in the year to March 2026
- **SUMCO** `Japan`. ¥409,670m of net sales in 2025

**China** China made about 93 percent of the world's polysilicon in 2023, and an estimated 98 percent of China's output was solar-grade. Eswin, mainland China's largest maker of 300 mm wafers, says it had about 7.7 percent of world 300 mm shipments in the first half of 2026.

**United States** None of the five largest wafer makers in 2025 was American. GlobalWafers of Taiwan won CHIPS awards of up to $406 million in December 2024 for wafer plants in Texas and Missouri.

Sources: chipsupplychain.org/silicon-and-wafers/

## 03 Chemicals, Gases and Photoresist

High concentration · Moderately hard to substitute

Japanese companies held 78 percent of world photoresist sales in 2023. Supply of chip chemicals, gases and metals has been disrupted three times since 2019.

- **JSR** `Japan`. A vehicle of the state-backed Japan Investment Corporation bought 84.36% of JSR in April 2024
- **Tokyo Ohka Kogyo** `Japan`. Resist maker with ¥139,668m of sales in the first half of fiscal 2026

**China** In March 2026 the Chinese resist maker Hubei Dinglong said China had yet to reach large-scale production of advanced resist for 248 nm, 193 nm or extreme-ultraviolet printing. Since August 2023 China has required licenses to export gallium and germanium.

**United States** In photoresist, American companies ranked a distant second to Japanese companies by sales in 2023. In 2021 JSR of Japan agreed to buy Inpria, an Oregon-based US maker of resist for extreme ultraviolet.

Sources: chipsupplychain.org/chemicals-gases-and-photoresist/

## 04 Lithography

**Extreme concentration** · Very hard to substitute

ASML, a Dutch firm, is the only maker of extreme ultraviolet scanners, the machines that print the finest layers of a chip. It sold 48 of them in 2025.

- **ASML** `Netherlands`. 100% of EUV (extreme ultraviolet) systems sold in 2025
- **Nikon** `Japan`. 22 new chip lithography systems in the year to March 2026

**China** In September 2025 CSIS judged that EUV remained firmly out of reach for China despite massive state investment. SMEE, a Chinese scanner maker, had its largest share in i-line tools, the coarsest kind, and even there held 4 percent of the world market as of 2025.

**United States** No American firm makes scanners, CSET found in 2021. But a US rule published in October 2023 covers immersion scanners with an alignment error between 1.5 nm and 2.4 nm bound for advanced chipmaking if they hold any American-made parts. The US entity list also requires a license to supply the firms it names.

Sources: chipsupplychain.org/lithography/

## 05 Photomasks and Pellicles

**Extreme concentration** · Hard to substitute

Only three firms in the world make the blank plates for the masks used to print the finest layers of leading-edge chips. Japan's AGC and Hoya are two of them.

- **AGC** `Japan`. Ranks itself No.2 worldwide in EUV mask blanks on its own fiscal 2026 estimates
- **Hoya** `Japan`. Said in its 2026 report that its share of mask blanks is exceptionally high

**China** As of 2026 SMIC runs two mask shops of its own in Shanghai. No Chinese firm appears in CSET's 2025 market shares for electron-beam mask writers, the machines that draw the pattern onto a mask.

**United States** In December 2025 Photronics called itself a leading worldwide maker of photomasks. As of 2026 KLA sells tools that inspect EUV masks and optical masks, the quartz masks used with deep ultraviolet light.

Sources: chipsupplychain.org/photomasks-and-pellicles/

## 06 Deposition and Etch

High concentration · Moderately hard to substitute

Deposition and etch tools build a chip layer by layer. China-based toolmakers' share of deposition tools rose from 2 to 7 percent between 2019 and 2024, by CSET's count.

- **Applied Materials** `United States`. 82.4% of physical vapor deposition tools in 2025, by CSET's count from TechInsights data
- **Lam Research** `United States`. 51.2% of conductor etch tools in 2025

**China** Chinese toolmakers held 12.4 percent of dry etch in 2025, according to CSET's analysis of TechInsights data. They held 2.6 percent of atomic layer deposition that year, by the same count.

**United States** A December 2024 US rule lists some dry etch, atomic layer deposition and deep-hole deposition tools as controlled exports. Its foreign direct product rule reaches some tools built outside the United States.

Sources: chipsupplychain.org/deposition-and-etch/

## 07 Metrology and Inspection

High concentration · Moderately hard to substitute

KLA, an American firm, held 56.8 percent of the market for tools that measure and inspect wafers in 2025, by CSET's count.

- **KLA** `United States`. 56.8% of process control tools in 2025 by CSET's count
- **Applied Materials** `United States`. 11.2% of process control tools in 2025

**China** Chinese firms held 0.7 percent of the world market for tools that measure and inspect wafers in 2025, by CSET's count. Skyverse, a Shenzhen maker of inspection and metrology tools, reported 2,053,298,171.74 yuan of revenue in 2025, up 48.75 percent on 2024.

**United States** In 2025 the American firm KLA was the world's largest maker of metrology and inspection tools. Citing Gartner, KLA puts its 2024 share at 56.5 percent, 6.5 times its nearest rival's share.

Sources: chipsupplychain.org/metrology-and-inspection/

## 08 Transistors and the Front End

**Extreme concentration** · Hard to substitute

Only TSMC, Intel and Samsung make 2 nm-class transistors in volume. TSMC plans A14 for 2028, with more than 20 percent higher logic density than its 2 nm-class N2.

- **TSMC** `Taiwan`. N2 in volume production since 4Q25, TSMC's first nanosheet node
- **Intel** `United States`. 18A in high-volume production in the United States, as of 2026

**China** SMIC makes chips on its N+3 process, which SemiAnalysis measured in June 2026 at 113.4 million transistors per square millimeter, against 107.7 million for TSMC's N6. It builds its tightest layers in several patterning steps because it cannot buy an extreme-ultraviolet scanner.

**United States** In August 2025 Intel called itself the only chipmaker that both develops and manufactures leading-edge logic chips in the United States. It says its 18A process brought gate-all-around transistors and backside power to market together in 2025.

Sources: chipsupplychain.org/transistors-and-front-end/

## 09 Foundries and Fabs

**Extreme concentration** · Hard to substitute

TSMC's six largest plants are all in Taiwan. In 2025 they could make more than 13 million wafers a year, out of more than 17 million for the whole company.

- **TSMC** `Taiwan`. 79% of its wafer revenue from 7 nm and below, 3Q26
- **Intel Foundry** `United States`. $5.8bn segment revenue in 2Q26, only $293m of it from outside customers

**China** Georgetown University's CSET reported in June 2024 that Huawei's Ascend 910B AI chip is believed to be made on SMIC's 7 nm-class N+2 process. But SemiAnalysis judged in April 2025 that most 910B and 910C chips were made at TSMC.

**United States** The US government agreed in August 2025 to buy a passive 9.9 percent stake in Intel, with no board seat. Announcing the deal, Intel called itself the only chipmaker that both develops and manufactures leading-edge logic chips in the United States. Intel Foundry lost $2.089 billion in the second quarter of 2026.

Sources: chipsupplychain.org/foundries-and-fabs/

## 10 Memory and HBM

**Extreme concentration** · Hard to substitute

SK hynix, Samsung and Micron make nearly all high bandwidth memory (HBM), the stacked memory beside AI chips. SK hynix took 63.2 percent of HBM revenue in 2025.

- **SK hynix** `South Korea`. 63.2% of HBM revenue in 2025
- **Samsung** `South Korea`. 19.3% of HBM revenue in 2025

**China** The Institute for Progress estimated in October 2025 that China would make virtually no HBM that year. Huawei said in September 2025 that its Ascend 950 chips would use two HBM designs of its own, without saying who would manufacture the memory.

**United States** Micron is the only American HBM maker. As of October 2026, License Exception HBM lets US and allied exporters ship HBM below 3.3 GB per second per square millimeter without a license, including to China when the chip designer buys it directly and it goes straight to the packaging site.

Sources: chipsupplychain.org/memory-and-hbm/

## 11 Advanced Packaging

**Extreme concentration** · Moderately hard to substitute

Epoch AI estimates that in 2025 Nvidia, Google, AMD and Amazon used around 90 percent of the capacity for CoWoS, TSMC's packaging for AI chips.

- **TSMC** `Taiwan`. CoWoS packaging used by Nvidia, Google, AMD and Amazon in 2025
- **Intel** `United States`. Foveros 3D packaging in volume at Rio Rancho, New Mexico, since January 2024

**China** Chinese firms had 5.3 percent of the market for fabrication tools built for advanced packaging in 2025, by CSET's analysis of TechInsights data. Between 2019 and 2024 they gained 5 percentage points of share in test tools but lost share in bonding and other conventional assembly and packaging tools.

**United States** Intel opened Fab 9 in Rio Rancho, New Mexico, in January 2024 to make its Foveros 3D packages in volume. Amkor broke ground in October 2025 on a $7B packaging and test campus in Peoria, Arizona, and expects production to start in early 2028.

Sources: chipsupplychain.org/advanced-packaging/

## 12 Substrates and PCBs

**Extreme concentration** · Hard to substitute

Ajinomoto, a Japanese food company, told investors in June 2026 that it holds over 95 percent of the market for the insulating film inside high-end chip packages.

- **Ajinomoto** `Japan`. Told investors in June 2026 that its build-up film has held over 95% of its market since launch
- **Nitto Boseki** `Japan`. Claimed an overwhelming advantage in low thermal expansion glass in its 2025 integrated report

**China** TTM, an American circuit board maker, said in its annual report for 2025 that its circuit board competitors are mostly based in China and Taiwan. Nitto Boseki, a Japanese glass fiber maker, said in its 2025 integrated report that Chinese and Taiwanese companies are aggressively catching up with it in the special glass sector.

**United States** The White House's 2021 supply chain review listed no American firm among the key substrate makers. In December 2024 the US Commerce Department finalized an award of up to $75 million to Absolics, an affiliate of Korea's SKC, for a glass substrate plant in Covington, Georgia.

Sources: chipsupplychain.org/substrates-and-pcbs/

## 13 Test and Assembly

High concentration · Moderately hard to substitute

Testing finds bad chips before they go into an expensive package. Japan's Advantest took about 65 percent of 2025 spending on chip testers, by its own estimate.

- **Advantest** `Japan`. About 65% of the tester market in 2025, by its own estimate
- **Teradyne** `United States`. $2.52bn of semiconductor test revenue in 2025, service included

**China** Georgetown's CSET, analyzing TechInsights data, finds that between 2019 and 2024 Chinese toolmakers gained 5 percentage points of world market share in test tools and 4 points in fabrication tools built for advanced packaging. But they lost share in bonding and other conventional assembly and packaging tools.

**United States** American firms held 23.5 percent of the market for test tools in 2025 and Japanese firms 63.0 percent, by CSET's analysis of TechInsights data. Amkor raised its planned investment in an Arizona packaging and test campus to $7 billion in October 2025. In June 2026 it announced a long-term partnership with TSMC, which will buy packaging and test services from it in Arizona.

Sources: chipsupplychain.org/test-and-assembly/

## 14 Systems and Networking

High concentration · Moderately hard to substitute

Nvidia's and AMD's racks wire 72 AI chips together to run as one computer. Many firms can assemble the racks, but lasers for the optical links were scarce in 2026.

- **Nvidia** `United States`. NVLink, InfiniBand and Spectrum-X Ethernet, which it credited for data center growth in the quarter to April 2026
- **Broadcom** `United States`. $16.7B of AI semiconductor revenue in the quarter to August 2026, up 221%

**China** Chinese firms such as InnoLight assemble high-speed optical modules in volume, at sites including Suzhou and, since 2019, Thailand. In February 2025 China put indium under new export restrictions. The same month it began requiring export permits for indium phosphide substrates, the wafers that lasers for optical links are grown on.

**United States** Coherent broke ground in Sherman, Texas, in June 2026 on an expansion of its factory there, which makes optical devices from indium phosphide. In August 2026 Nvidia said that Wistron's new plant in Fort Worth, Texas, was producing its GB300 superchips and that Foxconn was building a factory in Houston.

Sources: chipsupplychain.org/systems-and-networking/

## 15 Data Centers and Power

High concentration · Moderately hard to substitute

By mid-2026, companies had asked to connect about 474 GW of new demand to the Texas grid, mostly for data centers. Only about 205 GW of those requests already had the grid study needed to enter the grid operator's first review. The large transformers that connect such sites to the grid commonly took three years to arrive in 2024.

- **GE Vernova** `United States`. Gas equipment, transformers and switchgear, including through Prolec GE
- **Hitachi Energy** `Switzerland`. Power transformers and switchgear

**China** China used 25 percent of the world's data center electricity in 2024, second to the United States, by the International Energy Agency's estimate. Data centers made up around 3 percent of the growth in China's electricity demand from 2022 to 2024.

**United States** US factories supplied 29.2 percent of the large power transformers bought in the United States in 2023, measured by capacity. Constellation said in August 2026 that it expects to restart the Crane nuclear plant in 2027. Microsoft has agreed to buy the plant's power.

Sources: chipsupplychain.org/data-centers-and-power/

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Chip Supply Chain: How AI Chips Are Made, by Yashvardhan Sharma. Edition of 26 October 2026. The full chapters, with a source for every figure, are at chipsupplychain.org.

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